

Tanvir Ahmed

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LinkedIn • GitHub • ORCID

RESEARCH INTERESTS

Cyber Security and Intrusion Detection • On-Device LLM and Vision-Language Models • Trustworthy and Calibrated Machine Learning • Quantum Machine Learning • Medical Image Analysis

RESEARCH SUMMARY

I work on machine learning for network security and for medical diagnosis, and in both the thing that pulls me in is the rare class. In network traffic it is the attack type that appears a few hundred times among a million flows; on a sputum smear it is the one field a technician cannot afford to miss. Aggregate accuracy hides both. My ICTAI 2026 paper came out of exactly that: cut a feature set and the headline number barely moves while ransomware detection falls apart, so I built an operator that identifies which classes were harmed, traces the loss to the discarded features, and restores the smallest set that brings them back. The same question runs through the rest of my work: intrusion-detection feature selection written as a QUBO and solved with QAOA on IBM Quantum hardware against classical selectors on industrial IoT traffic (X-IIoTID, NF-ToN-IoT, NF-UNSW-NB15); a browser threat-detection agent that runs a vision-language model and an LLM entirely on the user's own machine; and two tuberculosis microscopy papers on deciding which detections a clinician should trust. I spend more time on evaluation than on models, because most results I have had to audit did not survive patient-disjoint splitting, a leakage check, or a look at per-class calibration. I would rather report parity than a number that will not hold.

EDUCATION

Independent University, Bangladesh

January 2015 - December 2019

Department of Computer Science and Engineering

Bachelor of Science in Computer Science

CGPA: 2.92 out of 4.00 (Final-year GPA: 3.07)

PUBLICATIONS

Conference papers

Accepted

- **Tanvir Ahmed**, Jannatul Ferdous Shorna, Md Mahbubur Akash, Md Khorshed Alam, and Md Tarek Hasan. "DARE-FS: A Selector-Agnostic Operator for Detecting and Repairing Class-Specific Feature-Selection Harm." *2026 IEEE 38th International Conference on Tools with Artificial Intelligence (ICTAI)*, 2026. Code: [GitHub].
- Md Faisal Sheikh, **Tanvir Ahmed**, Enam Ahmed Taufik, Abhijit Kumar Ghosh, Mohammad Zavid Parvez, Prabal Barua, and Subrata Chakraborty. "Beyond Confidence: A Leakage-Audited Selective-Prediction Study of Acid-Fast Bacilli Detection in Ziehl-Neelsen Microscopy." *2026 International Conference on Digital Image Computing: Techniques and Applications (DICTA)*, 2026. Code: [GitHub].
- Swagotam Malakar, **Tanvir Ahmed**, Abhijit Kumar Ghosh, Mohammad Zavid Parvez, Prabal Barua, and Subrata Chakraborty. "Towards Trustworthy Tuberculosis Detection: Subgroup Calibration of Object Detectors in Sputum-Smear Microscopy." *2026 International Conference on Digital Image Computing: Techniques and Applications (DICTA)*, 2026. Code: [GitHub].
- Dhiman Kumar Sarker, Fyaz Nafin Rahman, Md Rezaul Karim Khan, Sushmoy Paul Rohit, **Tanvir Ahmed**, and Md Mizanur Rahman. "Real-Time Browser Threat Detection Security Agent via Local VLM-LLM." *2026 5th IEEE International Conference on Robotics, Automation, Artificial Intelligence and Internet-of-Things (RAAICON)*, 2026. Code: [GitHub].

Journal paper

Under Review

- Abhijit Kumar Ghosh, Ar Rafiul Islam Rafim, Fahmid Al Farid, **Tanvir Ahmed**, Enam Ahmed Taufik, Jia Uddin, and Hezerul Abdul Karim. "Gated Dual-Scale Feature Fusion for Colorectal Cancer Grading in Centralized and Federated Pathology." *Informatics in Medicine Unlocked* (Elsevier), 2026. Code: [GitHub].

PROJECTS

QCIDS: Quantum-Classical Feature Selection for Network Intrusion Detection [GitHub]

- Cast intrusion-detection feature selection as a QUBO solved by QAOA ($p = 1, 2, 3$) on simulators and IBM 156-qubit hardware, by simulated annealing, and by classical baselines under matched leakage-free cross-validation; tightening the budget from $k = 20$ to $k = 15$ on NF-ToN-IoT cost greedy mutual information 4.3 macro-F1 points while QAOA $p = 2$ held at 0.763.

CovNCG: Training-Free Dense Quantum Feature Maps [GitHub]

- Designed a training-free quantum kernel from non-commuting generator blocks (three features per two qubits) with a closed-form kernel-variance result scaling as c^4 , which turns avoidance of exponential concentration into a tuning rule; evaluated with a projected kernel and SVM on five medical datasets, with IBM QPU validation.

Q-PlasmaNet: Hybrid Classical-Quantum Classifier for Plasma Cell Recognition [GitHub]

- Paired a frozen CNN backbone with a 4-8 qubit data re-uploading variational quantum head (PennyLane) at matched parameter budget, reaching F1 0.927 with 178 trainable parameters - statistically tied with a parameter-matched MLP (0.932) and below a 2.2M-parameter MobileNetV2 (0.949), establishing parity rather than quantum advantage.

PCMMD: Federated Learning and Knowledge Distillation for Plasma Cell Detection [GitHub]

- Benchmarked centralized, few-shot, knowledge-distillation (EfficientNet-B0 to MobileNetV3) and federated (FedAvg/FedProx, one client per patient) training for plasma-cell classification of bone-marrow crops, under patient-disjoint folds and image-hash de-duplication.

Direct-WAM: Self-Supervised Denoising for Two-Photon Calcium Imaging [GitHub]

- Proved a bias-floor theory of self-supervised objectives (per-objective error floor, ensemble bias-crossing, distillation transfer bound) and produced a corrected AI4Life CIDC25 benchmark in which a faithfully retrained DeepCAD-RT reaches +23.65 dB stSNR, while the F_0 -free Direct-WAM 3D U-Net distilled from a 13-way ensemble reaches +18.15 dB.

RESEARCH EXPERIENCE

Independent Researcher

Independent and collaborative research

Boca Raton, Florida, USA

January 2025 - Present

Research collaborators: Prof. Jia Uddin (Woosong University, South Korea); Dr. Mohammad Zavid Parvez (RMIT University, Australia); Assoc. Prof. Subrata Chakraborty (University of New England, Australia); Prof. Prabal Datta Barua (University of Southern Queensland, Australia)

- Work with international collaborators across machine learning for security, quantum optimization and medical imaging; five papers written in 2026, four accepted and one journal manuscript under review.
- Run the full experimental pipeline on each project: auditing the data for leakage, building and training the models, benchmarking against classical and quantum baselines, and reporting results over multiple seeds with confidence intervals.

PROFESSIONAL EXPERIENCE

Career Break - Relocation to the United States

F-2 dependent status

Boca Raton, Florida, USA

December 2022 - December 2024

- Relocated from Bangladesh to the United States as an F-2 dependent, a status that does not authorize employment; used the period for self-directed study in machine learning, which led into the independent research work above.

Upwork & Fiverr

Freelance Web and Software Developer

Remote - Dhaka, Bangladesh

December 2014 - November 2022

- Built and maintained websites and web applications for international clients, writing custom front-end and back-end code to each client's specification, and configuring and extending e-commerce platforms.
- Handled the ongoing side of the work: bug fixes, feature requests, data and inventory updates, and direct client support across an eight-year freelance practice.

IDOL Focus - A Professional Training Institute

Web Developer Intern

Dhaka, Bangladesh

September 2019 - December 2019

Supervisors: Md. Asif Bin Khaled, Lecturer, Department of Computer Science and Engineering, IUB (academic); Md. Al Mamun, IDOL Focus (industry)

- Built the Admin and Agent modules of an online job portal (*idolfocusjobs.com*) serving four user roles: role-based authentication, jobseeker and employer account management, CV database administration, job-posting approval, and payment handling.
- Implemented in PHP, JavaScript, Ajax, jQuery, and Bootstrap over MySQL; documented in an undergraduate internship report submitted toward the BSc degree.

TECHNICAL SKILLS

Security:	Network intrusion detection, IIoT traffic analysis, on-device threat detection agents
Trustworthy ML:	Calibration (ECE, Brier), selective prediction, subgroup and worst-group evaluation, leakage auditing
LLM & VLM:	Local LLM and vision-language model inference, 4-bit quantized deployment, multimodal threat detection
Programming:	Python, SQL, PHP, JavaScript
ML Frameworks:	PyTorch, Scikit-learn, LightGBM, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib
Quantum ML:	PennyLane, Qiskit, QAOA/QUBO formulation, variational circuits, quantum kernels, IBM Quantum hardware
Methods:	Object detection, federated learning, knowledge distillation, self-supervised learning, combinatorial optimization
Web:	PHP, MySQL, JavaScript, jQuery, Bootstrap, Shopify
Tools:	L ^A T _E X, Git, Jupyter Notebook, Google Colab
Languages:	Bangla (native), English (fluent)